

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & 0 \\ 0 & Q_{xx} & 0 \\ 0 & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)}$$

$$Q_{zz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & Q_{xxy} & 0 \\ 0 & -Q_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ Q_{xxy} & 0 & 0 \end{bmatrix}$$

$$Q_{xxy} = \mathbb{Q}_{3b}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ Q_{yzx} & 0 & 0 \\ 0 & -Q_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzx} = 3\mathbb{G}_u^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} Q_{yzx} & 0 & 0 \\ 0 & Q_{yzx} & 0 \\ 0 & 0 & Q_{xyz} \end{bmatrix}$$

$$Q_{yzx} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(2)}$$

$$Q_{xyz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & 0 & -Q_{xxzx} & 0 \\ Q_{xxzz} & Q_{xxzz} & Q_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & -Q_{xxzx} \\ Q_{xxzx} & -Q_{xxzx} & 0 & 0 & Q_{yzyz} & 0 \\ 0 & 0 & 0 & -Q_{xxzx} & 0 & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 3\mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)}$$

$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + \mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)}$$

$$Q_{xxzz} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)}$$

$$Q_{xxzx} = \mathbb{Q}_{4b}^{(1)}$$

$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 8\mathbb{Q}_{40}^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & Q_{xxzz} & 0 & Q_{xxzx} & 0 \\ 0 & 0 & Q_{xxzz} & 0 & -Q_{xxzx} & 0 \\ -Q_{xxzz} & -Q_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xxzx} \\ -Q_{xxzx} & Q_{xxzx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -Q_{xxzx} & 0 & 0 \end{bmatrix}$$

$$Q_{xxzz} = 3\mathbb{Q}_u^{(3)}$$

$$Q_{xxzx} = -2\mathbb{G}_{3b}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & 0 \\ 0 & Q_{yzyz} & 0 \\ 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)}$$

$$Q_{xyxy} = \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & Q_{xxzx} & 0 \\ 0 & -Q_{xxzx} & 0 \\ 0 & 0 & 0 \\ Q_{yzyz} & 0 & 0 \\ 0 & -Q_{yzyz} & 0 \\ Q_{xxzx} & 0 & 0 \end{bmatrix}$$

$$Q_{xxzx} = \mathbb{G}_{3b}^{(2)}$$

$$Q_{yzyz} = -3\mathbb{Q}_u^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} \\ Q_{yzxy} & -Q_{yzxy} & 0 & 0 & -Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzyz} = -3\mathbb{Q}_u^{(6)}$$

$$Q_{yzxy} = \mathbb{G}_{3b}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 & 0 & Q_{xxzx} & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & 0 & -Q_{xxzx} & 0 & 0 & -Q_{xxzx} & 0 \\ Q_{zzxx} & Q_{zzxx} & Q_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} & Q_{yzzy} & 0 & Q_{yzxy} \\ -Q_{yzxy} & Q_{yzxy} & 0 & 0 & Q_{yzzy} & 0 & 0 & Q_{yzyz} & 0 \\ 0 & 0 & 0 & -Q_{xxzx} & 0 & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} & -Q_{xxzx} & 0 & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 3\mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)}$$

$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + \mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)}$$

$$Q_{xxzz} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)}$$

$$Q_{xxzx} = -2\mathbb{G}_{3b}^{(1)} + \mathbb{G}_{3b}^{(2)} + \mathbb{Q}_{4b}^{(1)}$$

$$Q_{xxzx} = -2\mathbb{G}_{3b}^{(1)} - \mathbb{G}_{3b}^{(2)} + \mathbb{Q}_{4b}^{(1)}$$

$$Q_{zzxx} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)}$$

$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 8\mathbb{Q}_{40}^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)}$$

$$Q_{yzxy} = -2\mathbb{G}_{3b}^{(1)} - \mathbb{Q}_{4b}^{(1)}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} & Q_{yzyz} & 0 & Q_{yzxy} \\ Q_{yzxy} & -Q_{yzxy} & 0 & 0 & -Q_{yzyz} & 0 & 0 & -Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} & 0 & 0 & -Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)}$$

$$Q_{yzxy} = \mathbb{G}_{3b}^{(3)}$$

$$Q_{yzyz} = -\mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)}$$

$$Q_{xyxy} = \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & 0 & Q_{xxxz} \\ 0 & Q_{xxxx} & 0 \\ 0 & 0 & Q_{zzzz} \\ -Q_{xxxz} & 0 & Q_{yyyz} \\ Q_{zzxx} & 0 & 0 \\ 0 & Q_{xxyy} & 0 \\ 0 & Q_{yzyz} & 0 \\ Q_{xxzx} & 0 & -\frac{Q_{xxxx}}{2} + \frac{3Q_{xxyy}}{2} + Q_{yyyz} - Q_{yzyz} + Q_{zzxx} \\ \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} & 0 & -Q_{xxxz} \\ 0 & -Q_{xxxz} & 0 \end{bmatrix}$$

$$Q_{xxxx} = 3\mathbb{Q}_0^{(1)} + 3\mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)}$$

$$Q_{xxzx} = \mathbb{Q}_{4b}^{(1)}$$

$$Q_{zzzz} = 3\mathbb{Q}_0^{(1)} + 8\mathbb{Q}_{40}^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$

$$Q_{yyyz} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)}$$

$$Q_{zzxx} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)}$$

$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + \mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)}$$